GEODESIC STRING COUNTING INVARIANTS OF MANIFOLDS

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ABSTRACT. We study $\mathbb{Q}$-valued metric/topological invariants of manifolds, by counting closed geodesics, and using the Fuller index. One direct application is to geodesic counting is the following. For example we give conditions when a regular Finsler metric on T^n has at least 2 fixed class geodesic strings with multiplicity divisible by a fixed prime. We also give new conditions on when fibrations do not admit a forward complete metric with a unique and non-degenerate closed geodesic in a certain class. This partially, extends classical results of Preissmann. One of the main goals is to motivate a conjecture on full topological invariance of the geodesic count that we study. This conjecture has very strong consequences, and we partially verify it.

1. Introduction

Using counts of ***geodesic strings*** (equivalence classes of closed, constant speed geodesics up to reparametrization S^1 action), and the Fuller index [8], we will define certain rational number valued, deformation invariants for complete Riemann-Finsler manifolds. A basic case is a complete Riemannian manifold with non-positive sectional curvature, and in this case we get deformation invariants of the metric, provided the deformation is through non-positive sectional curvature metrics.

Studying connections of the Fuller index with Riemann-Finsler geometry was suggested by Fuller himself in the 1960's but was never much developed. An outline of Fuller's theory and basic definitions are given here in the Appendix.

Suppose for the moment X is compact. For a non-constant, free homotopy class β of a loop in X, we say that a metric g on X is ***β-regular*** if all of its class β closed geodesics are non-degenerate the usual sense, or equivalently the closed orbits of the associated geodesic flow are dynamically non-degenerate. For a β-regular g, with finitely many class β geodesic strings, we have the following formula for the count:

$$F(g,\beta) = \sum_o \frac{(-1)^{\mathrm{morse}(o)}}{\mathrm{mult}(o)} \in \mathbb{Q}, \tag{1.1}$$

where:

- The sum is over class β g-geodesic strings o.
- $\mathrm{morse}(o)$ denotes the Morse index of o, (meaning the Morse-Bott index of the associated critical submanifold (diffeomorphic to S^1) of the loop space).
- $\mathrm{mult}(o)$ is the geometric multiplicity, equivalently the order of the corresponding isotropy subgroup of S^1, where S^1 is acting on the loop space by reparametrization.

Let $\pi_1(X)$ denote the set of free homotopy classes of loops in X, and let $L_\beta X$ denote the subset of the free loop space with elements in class $\beta \in \pi(X)$. The following is a direct corollary of Theorem 1.24, and exemplifies the basic topological invariance of the count:

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Corollary 1.2. *Let X, g, β be as above and such that β is indivisible, (cannot be represented by a multiply covered loop). Then the S^1 equivariant homology $H_*^{S^1}(L_\beta X, \mathbb{Z})$ has finite total rank. Denote by $\chi^{S^1}(L_\beta X)$ the Euler characteristic of this homology. Then*

$$F(g, \beta) = \chi^{S^1}(L_\beta X).$$

Corollary 1.2 above, and its parent Theorem 1.24 lead us to conjecture full topological invariance, see Conjecture 1 in Section 1.1.1. The latter theorem is an important ingredient for some geometric applications concerning closed geodesic counts.

Assuming full topological invariance of our string counting invariant we get the following perhaps mysterious theorem.

Theorem 1.3 (Proof in Section 6)**.** *Assume the Conjecture 1, let $\beta \in \pi_1(T^n)$ be a non-trivial class and let g be a β-regular Finsler metric on T^n having finitely many β class geodesic strings. Suppose that o is a g-geodesic string s.t.:*

(1) *o has class β.*

(2) *$p|\operatorname{mult}(o)$, where p is prime,*

then there is at least one other such o (same p).

Remark 1.4. We can extend the above from metrics to Reeb vector fields on the unit cotangent bundle, representing the standard contact structure. But now counting closed Reeb orbits in classes $\widetilde{\beta}$, lifting a class β as above, (fix a unit speed parametrized representative and lift). However, we need a stronger form of the Conjecture 1, see Remark 1.17.

Without Conjecture 1 we have a local form of the above. Even this local form appears to be inaccessible to standard Morse theory techniques, is a new phenomenon and is likely the most central application of the paper. [1] The main ingredient for this is a product formula for the invariant F in the form of Theorem 8.1. The proof of the latter is rooted in dynamical ideas, but is mostly conceptual, sidestepping intricate analysis sometimes needed for infinite dimensional Morse theory.

Theorem 1.5 (Proof in Section 5)**.** *Let g be the standard flat metric on T^n, and $\beta \in \pi_1(T^n)$ a non-trivial class. For all C sufficiently large, there exists an $\epsilon > 0$ such that whenever g' is a Finsler metric C^2 ϵ close to g and is β-regular, the following holds. Suppose that o is a g'-geodesic string s.t.:*

(1) *The length $\ell_g(o) < C$.*

(2) *o has class β.*

(3) *$p|\operatorname{mult}(o)$, where p is prime,*

then there is at least one other such o (in particular same p). Moreover, any Finsler metric g_1 taut deformation equivalent to g (see Definition 1.11) has the same property as g, above. An example for such a metric is in Figure 1.

Recall that a now classical theorem of Preissmann [14] implies that there are no non-trivial compact products with a negative sectional curvature Riemannian metric. The following is one generalization and is one of the main concrete applications of the paper. We denote by $\pi_1^{inc}(X)$ the set of free homotopy classes of loops incompressible to the ends, in the sense of Definition 1.8, (non-constant classes when X is closed).

[1] Perhaps one needs something like infinite dimensional orbifold Morse homology, but this doesn't yet exist as far as I know.

Theorem 1.6 (Proof in Section 8)**.** *Let Y, Z be smooth manifolds, $\beta_Z \in \pi_1^{inc}(Z)$, $\chi(Y) \neq \pm 1$, Y is closed and admits a metric all of whose contractible geodesics are constant, e.g. a non-positive curvature Riemannian metric, Z admits a non-positive curvature metric. Then:*

- *X does not admit a complete Riemannian metric of negative sectional curvature.*
- *X does not admit a forward complete Finsler metric with a unique and non-degenerate class $i_*(\beta)$ where*
$$i_* : \pi_1^{\mathrm{inc}}(Z) \to \pi_1^{\mathrm{inc}}(X),$$
is induced by fiber inclusion $i : Z \to X$, for the projection $X \to Y$.

This theorem is very close to being sharp, for example the conclusion of the corollary is false if $Y = S^1$ and $Z = S^1 \times \mathbb{R}$. As $X = T^2 \times \mathbb{R}$ admits the warped product metric $g_{T^2} \times_{e^t} g_{\mathbb{R}}$ (with respect to the function $f = e^t$ on $\mathbb{R}$) where $g_{T^2}, g_{\mathbb{R}}$ are the flat metrics. This warped product has constant negative sectional curvature -1. So it is essential that not only $\pi_1(Z)$ be non-trivial but also that there is a class incompressible to the ends. Of course, $\chi(Y) = 1$ is also obviously essential, otherwise we may take $Y = pt$. The condition $\chi(Y) \neq -1$ is however not obviously essential.

Remark 1.7. The first part of the theorem above can be proved via the remarkable flat strip theorem [6].

We will also give various generalizations of the above results to fibrations. For fibrations, the most obvious analogue of Preissmann's theorem fails even assuming compactness. In fact, every closed 3-manifold X^3, for which there is no injection $\mathbb{Z}^2 \to \pi_1(X, x_0)$, and which fibers over a circle has a hyperbolic structure g_h, Thurston [16].

1.1. **Setup and more extensive statements.** Proofs of many results here are postponed till subsequent sections.

Terminology 1. From now on, all our metrics are Riemann-Finsler (a.k.a. Finsler) metrics unless specified to be Riemannian, and usually denoted by just g. Completeness, always means forward completeness, and it is an assumption for all our metrics. Curvature always means sectional curvature in the Riemannian case and flag curvature in the Finsler case. Thus we will usually just say complete metric g, for a forward complete Riemann-Finsler metric. A reader may certainly choose to interpret all metrics as Riemannian metrics, completeness as standard completeness, and curvature as sectional curvature.

In what follows $\pi_1(X)$ denotes the set of free homotopy classes of continuous maps $o : S^1 \to X$.

Definition 1.8. *Let X be a smooth manifold. Fix an exhaustion by nested compact sets $\bigcup_{i\in\mathbb{N}} K_i = X$, $K_i \supset K_{i-1}^\circ$ for all $i \geq 1$. We say that a class $\beta \in \pi_1(X)$ is end compressible if β is in the image of*
$$inc_* : \pi_1(X - K_i) \to \pi_1(X)$$
for all i, where $inc : X - K_i \to X$ is the inclusion map. We say that β is **end incompressible** *(or incompressible to the ends) if it is not end compressible.*

Let $\pi_1^{inc}(X)$ denote the set of such end incompressible classes. When X is compact, we set $\pi_1^{inc}(X) := \pi_1(X) - const$, where *const* denotes the set of homotopy classes of constant loops.

It is easily seen that the above is well defined (independent of the choice of an exhaustion) and moreover any homeomorphism $X_1 \to X_2$ of a pair of manifolds induces a set isomorphism $\pi_1^{inc}(X_1) \to \pi_1^{inc}(X_2)$.

Denote by $L_\beta X$ the class $\beta \in \pi_1^{inc}(X)$ component of the free loop space of X, with its compact open topology. Let g be a complete metric on X, and let $S(g, \beta) \subset L_\beta X$ denote the subspace of all constant speed parametrized, smooth, closed g-geodesics in class β.

Definition 1.9. *We say that a metric g on X is β-**taut** if it is complete and $S(g, \beta)$ is compact. We will say that g is **taut** if it is β-taut for each $\beta \in \pi_1^{inc}(X)$.*

Lemma 1.10. *A complete metric g with non-positive curvature satisfies:*

- *All of its closed geodesics are minimizing in their free homotopy class.*
- *It is taut.*

Proof. The first part is a standard consequence of the Cartan-Hadamard theorem. The second part follows by the first part and Lemma 4.1. □

It should be emphasized that taut metrics form a much larger class of metrics than just non-positive curvature metrics. For example any sufficiently C^1 small perturbation of a metric with non-positive curvature will be taut. (Indeed, this is crucial for the construction of our invariant.) Another class of examples comes by way of Lemma **??** ahead, these metrics may not be non-positively curved nor close to non-positively curved metrics. A simple concrete example very far away from being non-positively curves in Example (2).

Definition 1.11. *Let $\beta \in \pi_1^{inc}(X)$, and let g_0, g_1 be a pair of β-taut metrics on X. A β-**taut deformation** between g_0, g_1, is a continuous (in the topology of C^0 convergence on compact sets) family $\{g_t\}$, $t \in [0,1]$ of complete metrics on X, s.t.*

$$S(\{g_t\}, \beta) := \{(o,t) \in L_\beta X \times [0,1] \,|\, o \in S(g_t, \beta)\}$$

*is compact. We say that $\{g_t\}$ is a **taut deformation** if it is β-taut for each $\beta \in \pi_1^{inc}(X)$. The above definitions of tautness are extended naturally to the case of a smooth fibration $X \hookrightarrow P \to [0,1]$, with a smooth fiber-wise family of metrics.*

A useful criterion for β-tautness is the following.

Theorem 1.12 (Proof in Section 2.1)**.** *Let $\beta \in \pi_1^{inc}(X)$. Let $\{g_t\}_{t\in[0,1]}$ be a continuous family of complete metrics on X. Suppose that:*

$$\sup_t |\max_{o\in S(g_t,\beta)} \ell_{g_t}(o) - \min_{o\in S(g_t,\beta)} \ell_{g_t}(o)| < \infty,$$

where ℓ_{g_t} is the length functional with respect to g_t, then $\{g_t\}$ is β-taut.

For example, the hypothesis is trivially satisfied if g_t have the property that all their class β closed geodesics are minimal in their homotopy class.

Corollary 1.13. *If g_t, $t \in [0,1]$ have non-positive curvature then $\{g_t\}$ is taut.*

Proof. This follows by Theorem 1.12 and by Lemma 1.10. □

1.1.1. *The geodesic string counting invariant F.* Let $\mathcal{G}(X)$ be the set of equivalence classes of taut metrics g, where g_0 is equivalent to g_1 whenever there is a taut deformation between them. We may denote an equivalence class by its representative g by a slight abuse of notation. The following is proved in Section 4.

Theorem 1.14. *For each manifold X there is a natural, non-trivial functional:*

$$\mathrm{F} : \mathcal{G}(X) \times \pi_1^{inc}(X) \to \mathbb{Q},$$

extending the definition in (1.1).

For g β-regular and β-taut we have already defined $F(g, \beta)$ at the beginning of the Introduction. For a general β-taut g we define $F(g, \beta)$ in terms of the Fuller index.

Corollary 1.15. *Suppose for a pair g_0, g_1 of β-taut metrics on X:*

$$F(g_0, \beta) \neq F(g_1, \beta),$$

then any path $\{g_t\}$, connecting g_0, g_1, is not β-taut.

Proof. The fact that any connecting $\{g_t\}$ is not β-taut is just a direct corollary of the theorem above. □

I was unable to find such a pair g_0, g_1, in fact there is strong evidence to believe the following:

Conjecture 1. *F does not depend on the choice of a smooth structure and taut metric. In other words we have the following. Let X be a topological manifold, define:*

$$\mathcal{S}(X) : \pi_1^{inc}(X) \to \mathbb{Q} \sqcup \{\infty\}$$

by:

$$\mathcal{S}(X)(\beta) = \begin{cases} \infty, & \text{if } X \text{ does not admit a smooth structure and a } \beta\text{-taut Finsler metric.} \\ F(g, \beta), & \text{if } g \text{ is a } \beta\text{-taut Finsler metric on } X. \end{cases},$$

then $\mathcal{S}$ is well defined and hence determines a topological invariant of topological manifolds. So if $f : X_1 \to X_2$ is a homeomorphism then $\mathcal{S}(X_1)(\beta) = \mathcal{S}(X_2)(f_*(\beta))$.

Theorem 1.24 in the following section partially proves this conjecture.

Remark 1.16. The smooth manifold version of this conjecture, i.e. that $\mathcal{S}(X)$ is a smooth manifold invariant, is implied by the Reeb sky catastrophe conjecture described in the next section. However, the above seems to be much more basic as will be apparent from the proof of Theorem 1.24.

Remark 1.17. The formulation of the conjecture naturally extends to Reeb vector fields. That is let λ be a contact form on the unit cotangent bundle C of a smooth manifold X (for simplicity closed), with λ representing the Liouville contact structure (i.e. the standard contact structure). Assume that the space of closed, class $\widetilde{\beta}$ λ-Reeb orbits on C is compact. Then the Fuller index (See Section 4) of this compact set is a topological invariant of X.

1.1.2. *A short digression on sky catastrophes.* **??**.

Definition 1.18 (Preliminary)**.** *A* **sky catastrophe** *for a smooth family $\{X_t\}$, $t \in [0, 1]$, of non-vanishing vector fields on a closed manifold M is a continuous family of closed orbit strings $\tau \mapsto o_{t_\tau}$, o_{t_τ} is an orbit string of X_{t_τ}, $\tau \in [0, \infty)$, such that the period of o_{t_τ} is unbounded from above.*

Sky catastrophes first appeared in the work of Fuller [7]. One general definition appears in [15], which we review in the Appendix, the main point of this definition is that there are no regularity assumptions on $\{X_t\}$. The preliminary definition becomes equivalent to the more general definition given certain regularity conditions on the family $\{X_t\}$.

The following corollary of Theorem 1.12 may be of independent interest.

Corollary 1.19. *Sky catastrophes of vector fields on closed manifolds are not geodesible by metrics all of whose geodesics are minimal. That is if a family $\{X_t\}$ has a sky catastrophe, there is no family $\{g_t\}$ of metrics such that the orbits of X_t are unit speed parametrized g_t-geodesics.*

Fuller at the end of [8] has asked for any metric conditions on vector fields to rule out sky catastrophes. By the above, non-positivity of curvature is one such condition. So this is a partial answer to his question.

The main conjecture in [15] is that geodesible, and more generally Reeb sky catastrophes ($\{X_t\}$ are Reeb) are unstable, in the sense that there is a C^0 nearby family $\{X_t'\}$ which does not have a sky catastrophe (at least in the fixed class β). Let's call this ***Reeb sky catastrophe conjecture***. The reason to conjecture such a thing, is that a theorem in [15] says that if $\{o_{t_\tau}\}_\tau$ is a sky catastrophe for a family of Reeb vector fields $\{X_t\}$ then the length of the corresponding projection $\tau \mapsto t_\tau$ is infinite. (Assuming some minor regularity on $\{o_{t_\tau}\}$.) The latter suggests that the structure of such a sky catastrophe is very fragile.

Remark 1.20. The Reeb sky catastrophe conjecture implies the geodesible Seifert conjecture (apparently still an open problem), by the main result of [15]. Hence, this is a subtle question.

The following is a variant of Corollary 1.15.

Theorem 1.21. *Suppose for a pair g_0, g_1 of β-taut metrics on a closed manifold M:*

$$F(g_0, \beta) \neq F(g_1, \beta),$$

then any path $\{g_t\}$, connecting g_0, g_1, has a sky catastrophe, meaning that the corresponding family $\{X_t\}$ of Reeb vector fields on the unit cotangent bundle (cf. Section 4) has a sky catastrophe as in Definition C.1.

Proof. This directly follows by [15, Theorem 3.2]. ☐

So that if such a pair g_0, g_1 exists the Reeb sky catastrophe conjecture is disproved, and in a rather extreme fashion.

1.1.3. *Basic results on the invariant F.*

Definition 1.22. *Let $\beta \in \pi_1(X)$. For any based point $x_0 \in \operatorname{image} \beta \subset X$ (for $\operatorname{image} \beta$ the image of some representative of β) there is a naturally determined element $\beta_{x_0} \in \pi_1(X, x_0)$ well defined up to an inner automorphism, (concatenate a representative of β with a path from x_0 to a point in $\operatorname{image} \beta$).*

- *We say that a class $\beta \in \pi_1(X, x_0)$ is* **indivisible** *if whenever $\beta = \alpha^k$ for some $\alpha, k > 0$ then $k = 1$.*
- *We say that a class $\beta \in \pi_1(X, x_0)$ is a k-***power** *if $\beta = \alpha^k$, $k > 1$, for some α which is not a n-power for any n.*
- *We say that β is* **atomic** *if it is a k-power for some k.* [2]
- *We say that $\beta \in \pi_1(X)$ is indivisible, respectively is a k-power, respectively is atomic if for any x_0 as above, β_{x_0} is indivisible, respectively is a k-power, respectively is atomic. This notion does not depend on the choice of β_{x_0} in its inner automorphism class. In fact, β is k-power just means that β has a smooth representative with multiplicity k.*

Note that a class $\beta \in \pi_1^{inc}(X)$ is indivisible iff any representative of this class is not multiply covered.

[2] As we are working with non-compact manifold, we may in principle have infinitely divisible classes.

Example 1. Let g be a Riemannian metric with negative sectional curvature on a closed manifold X and $\beta \in \pi_1(X)$ a class represented by a multiplicity n closed geodesic, then

$$F(g, \beta) = \frac{1}{n}. \tag{1.23}$$

In particular, if β is indivisible then $F(g, \beta) = 1$. More generally, (1.23) holds whenever g has a unique and non-degenerate geodesic string in class β, where non-degenerate is as in the Introduction.

If $\beta \in \pi_1^{inc}(X)$ is indivisible, then it is easy to see that the reparametrization S^1 action on $L_\beta X$ is free (Section 7), so that $H_*^{S^1}(L_\beta X, \mathbb{Z}) \simeq H_*(L_\beta X/S^1, \mathbb{Z})$, where $H_*^{S^1}(L_\beta X, \mathbb{Z})$ denotes the S^1-equivariant homology. Moreover, we have the following result proved in Section 7.

Theorem 1.24. *Suppose that $\beta \in \pi_1^{inc}(X)$ is indivisible, and X admits a β-taut metric. Then $H_*^{S^1}(L_\beta X, \mathbb{Z})$ has finite total rank. Denote by $\chi^{S^1}(L_\beta X)$ the Euler characteristic of this homology. Then for any β-taut metric g on X:*

$$F(g, \beta) = \chi^{S^1}(L_\beta X).$$

Consequently, $F(g, \beta)$ is independent of the choice of β-taut metric g, for indivisible β. And Furthermore, Conjecture 1 holds on the subset of classes β which are indivisible.

Example 2. A basic example of a β-taut metric that is not nearby to a non-positive curvature metric is the following. Let g be a metric on the 2-torus with exactly two class β g-geodesic strings, for β one of the π_1 generators. We can take such a g_0 with arbitrarily large curvature, see Figure 1. The metric g_0 in this figure is moreover clearly taut deformation equivalent to the standard flat metric. Taking products of metrics of this sort gives higher dimensional examples.

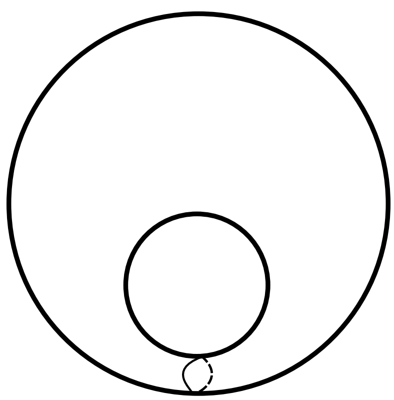

FIGURE 1. The class β is represented by the drawn curve.

More examples for the theorem above can be found by the proof of Theorem 1.26.

Remark 1.25. If β is a not indivisible, the idea behind Theorem 1.24 breaks down, as the S^1-equivariant homology of $L_\beta X$ may then be infinite dimensional even if X admits a β-taut g. As a trivial example, this homology is already infinite dimensional when g is negatively curved, and the class β geodesic is k-covered, as then this homology is the group homology of $\mathbb{Z}_k$. However for a general β we should still have that $F(g, \beta)$ is the orbifold Euler characteristic, of a suitable orbifold quotient $L_\beta X/S^1$ which in principle proves Conjecture 1. The difficulty in making this precise is that at the moment there is no such orbifold Euler characteristic theory as $L_\beta X/S^1$ is infinite dimensional (thinking of it as a stack or an orbifold).

We can use the above, and the product formula of Theorem 8.1 to get:

Theorem 1.26 (Proof in Section 8)**.** *Every rational number has the form $F(g, \beta)$ for some β-taut Riemannian g on some compact manifold X and for some β.*

1.1.4. *Applications to existence of negative curvature metrics.* A celebrated theorem of Preissmann [14] says that there are no negative sectional curvature metrics on compact products. Fibration counterexamples to Preissmann's product theorem certainly exist as mentioned in the Introduction. We are going to give a certain generalization of Preissmann's theorem to fibrations, with possibly non-compact fibers, also replacing the negative sectional curvature condition by a significantly weaker condition. Though the definitions may at first seem hard to understand, we just want enough generality to incorporate interesting "flat bundle" examples (with anti-Anosov or periodic holonomy behavior). The latter are of interest in 3-manifold theory, for instance vis-à-vis Thurston's classification [16].

Definition 1.27. *For a fiber bundle $Z \hookrightarrow X \to Y$, we say that $\beta \in \pi_1(X)$ is a* **fiber class** *if it is in the image of the set map $i_Z : \pi_1(Z) \to \pi_1(X)$, induced by inclusion.*

Definition 1.28. *Let*
$$Z \hookrightarrow X \xrightarrow{p} Y$$
be a smooth fiber bundle, with Y connected. Fix $y_0 \in Y$, and let H_{y_0} denote the holonomy group acting on $\pi_1^{\mathrm{inc}}(Z_{y_0})$. For
$$\beta \in \pi_1^{\mathrm{inc}}(Z_{y_0}),$$
*a fiber class, we say that p is β-***holonomy-finite** *if the holonomy orbit*
$$H_{y_0} \cdot \beta_Z$$
is finite, for any β_Z, such that $(i_Z)_*(\beta_Z) = \beta$. (Note that the action of H_{y_0} s necessarily transitive so that this is independent of the choice of β_Z.)

This condition is independent of the choice of y_0, up to the natural identification of holonomy actions at different base points.

Definition 1.29. *Let*
$$Z \hookrightarrow X \xrightarrow{p} Y$$
be a smooth fiber bundle, and let g be a Riemannian metric on X. Let
$$T^{\mathrm{vert}} X = \ker(dp)$$
and let
$$\pi^{\mathrm{vert}} : TX \to T^{\mathrm{vert}} X$$
*be the g-orthogonal projection. We say that p is g-***parallel** *if*
$$\nabla \pi^{\mathrm{vert}} = 0.$$
Equivalently,
$$\nabla_A (\pi^{\mathrm{vert}} B) = \pi^{\mathrm{vert}} (\nabla_A B)$$
for all vector fields A, B on X.

Definition 1.30. *Let β be a fiber class, for a smooth fibration*
$$p : Z \hookrightarrow (X, g) \to (Y, g_Y).$$
*We say that this is a β-***periodic fibration** *if:*

(1) *g is a complete Riemannian metric.*

(2) *p is g-parallel and is a Riemannian submersion.*

(3) *p is β-holonomy-finite.*

(4) *the fiber metrics g_y on Z_y are taut.*

The metrics g, g_Y may be kept implicit. In the above definition of a β_Z-periodic fibration and the hypothesis of the following theorem we need the auxiliary metric g on X to be Riemannian, and there is no obvious extension of the theorem to the Riemann-Finsler case. However, the conclusions of the theorem are for Riemann-Finsler metrics. The following is perhaps the main technical result of the paper, proved in Section 8.

Theorem 1.31. *Let $p : Z \hookrightarrow (X, g) \to (Y, g_Y)$ be a β-periodic fibration, where $\beta \in \pi_1^{inc}(X)$ is a fiber class. Suppose further that Y is connected, closed, $\chi(Y) \neq \pm 1$ and is such that all smooth closed contractible g_Y-geodesics in Y are constant. Then the following holds:*

- *X does not admit a complete Riemannian metric with negative curvature.*
- *X does not admit a forward complete Riemann-Finsler metric with a unique and non-degenerate class β geodesic string.*

Note that $\chi(Y) \neq 1$ is of course essential, as the trivial fibration $X \to \{pt\}$, with X admitting a complete negatively curved metric, will satisfy the hypothesis. The condition that there is a fiber class $\beta \in \pi_1^{inc}(X)$ is also essential, for any vector bundle over a manifold admitting a Riemannian metric of negative curvature admits a metric of negative curvature, Anderson [1].

Theorem 1.6 gives one set of examples. A further basic set of examples for the theorem is obtained by starting with a Riemannian manifold (Y, g_Y), and any homomorphism

$$\phi : \pi_1(Y, y_0) \to \mathrm{Isom}(Z, g_Z), \quad \text{(the group of all isometries).} \tag{1.32}$$

where g_Z is a taut Riemannian metric and $\beta_Z \in \pi_1^{inc}(Z)$. (For example (Z, g_Z) is a non-simply connected complete hyperbolic surface of genus at least one). Suppose further:

(1) The orbit
$$Orb := \bigcup_{\gamma \in \pi_1(Y, y_0)} \phi_*(\gamma)(\beta_Z)$$
is finite.

(2) Y is closed and connected.

(3) All contractible smooth closed g_Y geodesics in Y are constant.

We have the obvious induced diagonal action

$$\pi_1(Y, y_0) \to \mathrm{Diff}(Z \times \widetilde{Y}), \text{ (the group of all diffeomorphisms),}$$
$$\gamma \mapsto ((z, y) \mapsto (\phi(\gamma)(z), \gamma \cdot y)),$$

for $\widetilde{Y}$ the universal cover of Y. Taking the quotient of $Z \times \widetilde{Y}$ by this action, we get an associated "flat" bundle $Z \hookrightarrow X_\phi \xrightarrow{p} Y$, with a metric g_ϕ induced from the product metric $\widetilde{g} = g_Z \oplus g_Y$, on the covering space $q : Z \times \widetilde{Y} \to Z \times Y$. The natural projection $p : X_\phi \to Y$ is then a Riemannian submersion. In particular p is a β-periodic submersion. Hence $p : (X_\phi, g_\phi) \to (Y, g_Y)$ satisfies the hypothesis of Theorem 1.31. Yet more concretely:

Example 3. Suppose we have $\beta_Z \in \pi_1^{inc}(Z)$, and let $\phi : Z \to Z$ be an isometry of a taut metric g_Z, such that Orb is finite. Then for $\beta = (i_Z)_*(\beta_Z)$, by the construction above, the mapping torus

$$(Z, g_Z) \hookrightarrow (X_\phi, g_\phi) \xrightarrow{\pi} S^1$$

has the structure of a β-periodic fibration, satisfying the hypothesis of Theorem 1.31.

Corollary 1.33. *Let*
$$(Z_{g,Z}) \hookrightarrow (X_\phi, g_\phi) \to (Y, g_Y)$$
be as in the construction above for Z, g_Z having non-positive curvature, and such that Orb is finite. Let $\beta_Z \in \pi_1^{inc}(Z)$. Then if $\chi(Y) \neq \pm 1$:

1. *X_ϕ does not admit a complete Riemannian metric with negative curvature.*
2. *Moreover, X_ϕ does not admit a Riemann-Finsler metric with a unique and non-degenerate class β geodesic string, for $\beta = i_*(\beta_Z)$ as above.*

As a special case, this applies to the mapping tori X_ϕ, for $\phi : Z \to Z$ an isometry of a complete Riemannian non-positively curved metric on Z, satisfying the finiteness condition 1. (The non-positive curvature hypothesis is for concreteness we may of course replace this condition by tautness.)

In the special case when Z is compact, the first part of the above corollary can likely be deduced, from Preissmann's theorem, see also [3, Theorem 9.3.4] for a generalization that fits our Finsler setting. The second part is new even when Z is compact.

The following questions seem to be open.

Question 1. Is the β-holonomy finite assumption necessary for Corollary 1.33? Note that the β-holonomy finite assumption holds automatically if Z is closed since the isometry group is compact.

Question 2. Let Z be an infinite type surface, can (X_ϕ, g_ϕ) have a complete negative curvature metric? The corollary above cannot give a negative answer since in this case $X_\phi \to S^1$ may not be β-holonomy finite. But the answer is very likely no, as the intuition of Thurston-Nielsen classification (it cannot be directly applied in the infinite type setting) suggests that ϕ should have pseudo-Anosov type behavior which an isometry even on an infinite type surface cannot have.

2. Compactness preliminaries

Lemma 2.1. *Suppose that g is a complete metric on X, $\beta \in \pi_1^{inc}(X)$ and let $S \subset L_\beta X$ be a subset on which the g-length functional is bounded from above. Then the images in X of elements of S are contained in a fixed compact subset of X.*

Proof. Suppose otherwise. Fix an exhaustion by nested compact sets
$$\bigcup_{i \in \mathbb{N}} K_i = X, \quad K_i^\circ \supset K_{i-1}.$$
Then either there is sequence $\{o_i\}_{i\in\mathbb{N}}$, $o_i \in S$ s.t. $o_i \in K_i^c$, for K_i^c the complement of K_i, which contradicts the fact that β is end incompressible. Or there is a sequence $\{o_k\}_{k\in\mathbb{N}}$, $o_k \in S$ s.t.:

1. Each o_k intersects K_{i_0} for some i_0 fixed.
2. For each $i \in \mathbb{N}$ there is a $k_i > i$ s.t. o_{k_i} is not contained in K_i.

Now if $\mathrm{diam}(o_k)$ is bounded in k, then condition 1 implies that o_k are contained in a set of bounded diameter. (Here $\mathrm{diam}(o_k)$ denotes the diameter of image o_k.) Consequently, by Hopf-Rinow theorem [2], o_k are contained in a compact set. But this contradicts condition 2, and the fact that K_i form an exhaustion of X.

Thus, we conclude that $\mathrm{diam}(o_k)$ is unbounded, but this contradicts the hypothesis. □

Lemma 2.2. *Let g be a complete metric on X, and let $\beta \in \pi_1^{\mathrm{inc}}(X)$. Denote by $S_A(g, \beta) \subset S(g, \beta)$ the subset of elements with g-length at most A. Then $S_A(g, \beta) \subset S(g, \beta)$ is compact.*

Proof. By Lemma 2.1, the images of elements of $S_A(g,\beta) \subset S(g,\beta)$ are contained in compact K. Since elements of $S_A(g,\beta) \subset S(g,\beta)$ are parametrized by constant speed, and since we have a length bound $S_A(g,\beta)$ is an equicontinuous collection. The result then follows by the Arzelà-Ascoli theorem. (This uses the standard fact that C^0 convergence of closed geodesics upgrades to C^∞ convergence provided g is smooth.) □

Lemma 2.3. *Let g be a complete metric on X, and let $\beta \in \pi_1^{\mathrm{inc}}(X)$. Then there exists a closed g-geodesic minimizing length among all loops in the free homotopy class β.*

Proof. Suppose otherwise and let
$$A = \inf_{\gamma \in L_\beta X} \ell_g(\gamma).$$
By the hypothesis that there is no geodesic with length A, by Lemma 2.2, there is an $\epsilon > 0$ such that there is no closed geodesic o with $A < \ell_g(o) < A + \epsilon$.

By Lemma 2.1 the subset $M_{\beta,A} \subset L_\beta X$ consisting of elements with length less than $A+\epsilon$ is contained in a compact K. Pick $o \in M_{\beta,A}$ and apply the curve shortening process to it. Let o_k, $k \in \mathbb{N}$ be the stages of this process. Then $\operatorname{image} o_k \subset K$. In particular $\{o_k\}$ must converge to a geodesic, but this contradicts the condition on ϵ. □

2.1. **Proof of Theorem 1.12.** Let $\{g_t\}$, $t \in [0,1]$ be as in the hypothesis, with
$$\sup_t |\max_{o \in S(g_t,\beta)} \ell_{g_t}(o) - \min_{o \in S(g_t,\beta)} \ell_{g_t}(o)| < C, \tag{2.4}$$
and suppose that
$$\sup_{(o,t) \in \mathcal{O}(\{g_t\},\beta)} \ell_{g_t}(o) = \infty.$$
Then we have a sequence $\{o_k\}$, $k \in \mathbb{N}$, of closed class β g_{t_k}-geodesics in X, satisfying:

(1) $\lim_{k\to\infty} t_k = t_\infty \in [0,1]$.

(2) $\lim_{k\to\infty} \ell_{g_{t_k}}(o_k) = \infty$, where $\ell_{g_{t_k}}(o_{t_k})$ is the length with respect to g_{t_k}.

Applying Lemma 2.3, we may choose a length-minimizing closed g_{t_∞}-geodesic o_∞ in the class β. And let L denote its length g_∞ length. Let g_{aux} be a fixed auxiliary metric on X, and let L_{aux} be the g_{aux} length of o_∞.

Define a pseudo-metric d_{C_0} on the space of metrics on X as follows. Set $K = \operatorname{image} o_\infty$. And set
$$V \subset TX = \{v \in TX \,|\, \pi(v) \in K \text{ for } \pi : TX \to X \text{ the canonical projection, and } |v|_{aux} = 1\},$$
where $|v|_{aux}$ is the norm taken with respect to g_{aux}.

Then define:
$$d_{C^0}(g_1,g_2) = \sup_{v\in V} ||v|_{g_1} - |v|_{g_2}|.$$

By Properties 1 and 2 we may find a $k > 0$ such that:
$$d_{C^0}(g_{t_k}, g_{t_\infty}) < \epsilon \tag{2.5}$$
and
$$\ell_{g_{t_k}}(o_k) > C + L + L_{aux} \cdot \epsilon. \tag{2.6}$$

By (2.5), we have:
$$\ell_{g_{t_k}}(o_\infty) < \ell_{g_{t_\infty}}(o_\infty) + L_{aux} \cdot \epsilon = L + L_{aux} \cdot \epsilon.$$
Combining with (2.6) we get:
$$\ell_{g_{t_k}}(o_k) > \ell_{g_{t_k}}(o_\infty) + C.$$

Since we may find a closed g_{t_k}-geodesic o' satisfying $\ell_{g_{t_k}}(o') \leq \ell_{g_{t_k}}(o_\infty)$, we get that

$$|\max_{o \in S(g_{t_k}, \beta)} \ell_{g_{t_k}}(o) - \min_{o \in S(g_{t_k}, \beta)} \ell_{g_{t_k}}(o)| > C,$$

and so we are in contradiction.

Thus,

$$\sup_{(o,t) \in \mathcal{O}(\{g_t\}, \beta)} \ell_{g_t}(o) < \infty.$$

Then compactness of $\mathcal{O}(\{g_t\}, \beta)$ follows by a direct analogue of Lemma 2.2. □

3. Preliminaries on Reeb flow

Let (C^{2n+1}, λ) be a contact manifold with λ a contact form, that is a one form s.t. $\lambda \wedge (d\lambda)^n \neq 0$. Denote by R^λ the Reeb vector field satisfying:

$$d\lambda(R^\lambda, \cdot) = 0, \quad \lambda(R^\lambda) = 1.$$

Recall that a ***closed λ-Reeb orbit*** (or just Reeb orbit when λ is implicit) is a smooth map

$$o : (S^1 = \mathbb{R}/\mathbb{Z}) \to C$$

such that

$$\dot{o}(t) = cR^\lambda(o(t)),$$

with $\dot{o}(t)$ denoting the time derivative, for some $c > 0$ called period. Let $S(R^\lambda, \beta)$ denote the space of all closed λ-Reeb orbits in free homotopy class β, with its compact open topology. And set

$$\mathcal{O}(R^\lambda, \beta) = S(R^\lambda, \beta)/S^1,$$

where $S^1 = \mathbb{R}/\mathbb{Z}$ acts by reparametrization $t \cdot o(\tau) = o(t + \tau)$.

4. Definition of the functional F and proofs of auxiliary results

Let X be a manifold with a taut metric g. Let C be the unit cotangent bundle of X, with its Liouville contact 1-form λ_g. If $o : S^1 = \mathbb{R}/\mathbb{Z} \to X$ is a constant speed closed geodesic, it has a canonical lift $\widetilde{o} : S^1 \to C$, which is a closed flow line of $s \cdot R^{\lambda_g}$, where $s = |\frac{do}{dt}|$ is the speed of the geodesic, i.e. it is a Reeb orbit.

If $\beta \in \pi_1^{inc}(X)$, let $\widetilde{\beta} \in \pi_1(C)$ denote class $[\widetilde{o}] \in \pi_1(C)$, where o is a constant speed closed geodesic representing β.

Let $S(R^{\lambda_g}, \widetilde{\beta})$ be the orbit space as in Section 3, for the Reeb flow of the contact form λ_g. And set

$$\mathcal{O}_{g,\beta} = \mathcal{O}(R^{\lambda_g}, \widetilde{\beta}) := S(R^{\lambda_g}, \widetilde{\beta})/S^1,$$

which by construction is identified with the space of class β g-geodesic strings. By the tautness assumptions $\mathcal{O}_{g,\beta}$ is compact.

We then define

$$F(g, \beta) = i(\mathcal{O}_{g,\beta}, R^{\lambda_g}, \widetilde{\beta}) \in \mathbb{Q}$$

where the right hand side is the Fuller index as outlined in the Appendix ??. As a basic example we have:

Lemma 4.1. *Suppose that g is a complete metric on X, all of whose class $\beta \in \pi_1^{inc}(X)$ geodesics are minimal, then g is β-taut.*

Proof. This is a corollary of Lemma 2.2. □

Proof Theorem 1.14. Let $\beta \in \pi_1^{inc}(X)$, be given and let g be β-taut. We just need to prove that $F(g, \beta)$ is invariant under a β-taut deformation of g. So let $\{g_t\}$, $t \in [0,1]$ be a β-taut deformation of metrics on a compact manifold X. Let $R^{\lambda_{g_t}}$ be the geodesic flow on the g_t unit cotangent bundle C_t. Trivializing the family $\{C_t\}$ we get a family $\{R_t\}$ of flows on $C \simeq C_t$, with R_t conjugate to $R^{\lambda_{g_t}}$.

Let $\mathcal{O}(\{R_t\}, \widetilde{\beta})$ be the cobordism as in (A.1), where $\widetilde{\beta} \in \pi_1(C)$ is as above. Then $\mathcal{O}(\{R_t\}, \widetilde{\beta})$ is compact as $S(\{g_t\}, \beta)$ is compact by assumption. Basic invariance of the Fuller index, that is (A.2), immediately yields: $F(g_0, \beta) = F(g_1, \beta)$. □

5. Proof of Theorem 1.5

Let β be a non-zero class, as in the statement. Let $Y \subset T^n$ be a totally g-geodesic submanifold diffeomorphic to T^{n-1}, containing image β. Let $\alpha \in H^1(T^n, \mathbb{Z})$ be the Poincare dual of Y and let $p : T^n \to S^1$ be the classifying map of α. Then clearly p is a β-taut fibration with respect to g.

Let $C > \max_{\gamma \in [\beta]} \ell_g(\gamma)$ be given. Set

$$U_C = \{\gamma \in \mathcal{L}_{\widetilde{\beta}} S^* T^n \mid \ell g_S(\gamma) < C\}.$$

Let W corresponding to U_C be as in the isolation Lemma A.3 and let ϵ be the Fuller constant of N_0, U_C, W, where $N_0 = S(R_g^\lambda, \beta)$.

If g' is C^2 ϵ-close to g, then applying Lemma A.4 we get:

$$F(g, \beta) := i(N_0, \widetilde{\beta}) = i(N_1, \widetilde{\beta}), \tag{5.1}$$

where $N_1 = U_C \cap \mathcal{O}(R^{\lambda_{g'}}, \widetilde{\beta})$.

By Theorem 1.24, $F(g, \beta) = 0$, since $\chi(S^1) = 0$. Also the elements of N_1 are in correspondence with elements of

$$\mathcal{O}_C(g', \beta) = \{o \in \mathcal{O}(g', \beta) \,|\, \ell_g(o) < C\}.$$

We may thus rewrite (5.1) as:

$$\sum_{o \in \mathcal{O}_C(g', \beta)} \frac{(-1)^{\mathrm{morse}(o)}}{\mathrm{mult}(o)} = 0, \tag{5.2}$$

To finish the argument, enumerate elements of $\mathcal{O}_C(g', \beta)$ as $o_1, \ldots, o_n$ we then get:

$$\sum_i (-1)^{\mathrm{morse}(o_i)} \prod_{j \neq i} \mathrm{mult}(o_j) = 0.$$

We show that if a prime p divides the multiplicity $\mathrm{mult}(o_k)$ for some $k \in \{1, \ldots, n\}$, then there exists at least one other index $m \neq k$ such that p divides $\mathrm{mult}(o_m)$. This will complete the proof.

Let $M_i = \prod_{j \neq i} \mathrm{mult}(o_j)$ and let $\sigma_i = (-1)^{\mathrm{morse}(o_i)}$. The original equation is:

$$\sum_{i=1}^n \sigma_i M_i = 0.$$

Suppose $p \mid \mathrm{mult}(o_k)$. Then

$$\sum_{i=1}^n \sigma_i M_i \equiv 0 \pmod{p}. \tag{5.3}$$

For any index $i \neq k$, the product M_i contains the factor $\mathrm{mult}(o_k)$:

$$M_i = \mathrm{mult}(o_k) \cdot \prod_{j \neq i,k} \mathrm{mult}(o_j).$$

Since $p \mid \mathrm{mult}(o_k)$, it follows from (5.3) that:

$$\sigma_k M_k + 0 \equiv 0 \pmod{p}.$$

And so

$$M_k \equiv 0 \pmod{p}.$$

By definition, $M_k = \prod_{j \neq k} \mathrm{mult}(o_j)$. Since p is prime and $p \mid \prod_{j \neq k} \mathrm{mult}(o_j)$, by Euclid's Lemma,

$$p \mid \mathrm{mult}(o_j) \text{ for some } j \neq k.$$

And this completes the proof of the claim and the main theorem.

□

6. Proof of Theorem 1.3

Let $p : T^n \to S^1$ be the β-taut fibration with respect to g_0, as in the proof of Theorem 1.5 above. By Theorem 1.24, $F(g_0, \beta) = 0$. Hence, as we assumed Conjecture 1, for any other β-regular g on M with finitely β-class geodesic strings we have:

$$\sum_{o \in \mathcal{O}(g,\beta)} \frac{(-1)^{\mathrm{morse}(o)}}{\mathrm{mult}(o)} = 0.$$

Then the conclusion of the Theorem follows by the same arithmetic argument as in the proof of Theorem 1.5 above. □

7. Proof of Theorem 1.24

For simplicity, as the full argument is already lengthy, we prove the theorem under the assumption that g is Riemannian. In the Finsler case we must use a more modern variational setup for the Hilbert manifold of H^1-loops. In the Finsler case the energy functional is generally only C^1, rather than C^2, on the Hilbert loop space. Nevertheless the required Palais–Smale compactness and critical point theory for closed Finsler geodesics are standard; see Caponio–Javaloyes–Masiello [4], Mercuri [11], and Lu [10]. With these replacements, the compactness argument for end-incompressible classes, and the Fuller-index invariance argument go through without substantial change.

The circle S^1 acts on $L_\beta X$ by reparametrization:

$$(s \cdot \gamma)(t) = \gamma(t + s).$$

If some $\gamma \in L_\beta X$ had nontrivial stabilizer, then γ would factor through a nontrivial covering $S^1 \to S^1$, contradicting indivisibility. Therefore the S^1-action on $L_\beta X$ is free. Hence

$$H_*^{S^1}(L_\beta X, \mathbb{Z}) \cong H_*(L_\beta X / S^1, \mathbb{Z}),$$

and we will construct a finite total rank model for the latter.

Let

$$\mathcal{L}_\beta X$$

denote the Hilbert manifold of H^1-loops in the free homotopy class β.

Consider the energy functional

$$\mathcal{E}_g : \mathcal{L}_\beta X \to \mathbb{R}, \qquad \mathcal{E}_g(\gamma) = \int_{S^1} |\dot{\gamma}(t)|_g^2 \, dt.$$

Since $S(g, \beta)$ is compact, $\mathcal{E}_g$ is bounded above on the critical set of $\mathcal{E}_g$ in class β. Choose

$$E_0 > \max_{\gamma \in S(g,\beta)} \mathcal{E}_g(\gamma) = M^2,$$

which exists by the β-tautness assumption. Set

$$U_{E_0} = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_g(\gamma) < E_0\}.$$

Lemma 7.1. *The inclusion*

$$U_{E_0} \hookrightarrow \mathcal{L}_\beta X$$

is an S^1-equivariant homotopy equivalence, and

$$U_{E_0}/S^1 \simeq L_\beta X/S^1.$$

Proof. The energy functional $\mathcal{E}_g$ is smooth, S^1-invariant, satisfies the Palais–Smale condition, and its negative gradient flow (using the canonical g-induced S^1-equivariant Hilbert metric) has the usual compactness properties. Indeed, let $\{\gamma_j\} \subset \mathcal{L}_\beta X$ be a Palais–Smale sequence for $\mathcal{E}_g$ with

$$\mathcal{E}_g(\gamma_j) \leq A.$$

The energy bound gives a uniform length bound:

$$\ell_g(\gamma_j) \leq \mathcal{E}_g(\gamma_j)^{1/2} \leq A^{1/2}.$$

Since $\beta \in \pi_1^{\mathrm{inc}}(X)$, Lemma 2.1 implies that all images $\gamma_j(S^1)$ lie in a fixed compact subset $K \subset X$.

On this compact set, the metric g is uniformly equivalent to the Euclidean metric in finitely many coordinate charts, and the Christoffel symbols are uniformly bounded. Hence the energy bound gives a uniform H^1-bound. After passing to a subsequence, we have

$$\gamma_j \to \gamma_\infty$$

uniformly and weakly in H^1. The Palais–Smale condition

$$\|d\mathcal{E}_g(\gamma_j)\| \to 0$$

then gives the usual strong H^1-compactness for the geodesic energy functional. Thus $\mathcal{E}_g$ satisfies the Palais–Smale condition on every energy sublevel of $\mathcal{L}_\beta X$.

By the choice of E_0, the functional $\mathcal{E}_g$ has no critical points in

$$\{\mathcal{E}_g \geq E_0\}.$$

Equivalently, for every $A > E_0$, it has no critical points in the closed region

$$\{E_0 \leq \mathcal{E}_g \leq A\}.$$

The standard deformation lemma Palais [13, Section 10, Proposition (2)] therefore implies that

$$\{\mathcal{E}_g < E_0\} \hookrightarrow \{\mathcal{E}_g < A\}$$

is an S^1-equivariant homotopy equivalence for every $A > E_0$. Taking the direct limit over A, we obtain an S^1-equivariant homotopy equivalence

$$U_{E_0} \simeq \mathcal{L}_\beta X.$$

Since $\mathcal{L}_\beta X$ is S^1-equivariantly homotopy equivalent to $L_\beta X$, this also gives an S^1-equivariant homotopy equivalence:

$$U_{E_0} \simeq L_\beta X,$$

where by slight abuse U_{E_0} also denotes the E_0 sublevel set in the smooth loop space $L_\beta X$. In particular,

$$U_{E_0}/S^1 \simeq L_\beta X/S^1.$$

□

We now have to do some manipulations for the Fuller index calculations. The intricacy has to do with the fact that we need to perturb g, but the perturbed metric g' may now have a non-compact set $S(g', \beta)$. Let g_S denote the Sasaki metric on $C = S^*X$ induced by g as in Example 4 in the Appendix. Let

$$\widetilde{U}_{\sqrt{E_0}} \subset L_{\widetilde{\beta}} S^* X$$

be the subset of loops γ satisfying $\ell_{g_S}(\gamma) < \sqrt{E}_0$. Then (A.5) and the defining property of E_0 readily imply that $\widetilde{U}_{\sqrt{E_0}}$ is an isolating neighborhood of $N_0 = O(R_g^{\lambda}, \widetilde{\beta})$ in the sense of the isolation Lemma A.3, and is trivially g_S-length bounded. Let W be as in Lemma A.3.

Let ϵ be the corresponding Fuller constant of $N_0, \widetilde{U}_{\sqrt{E_0}}, W$ as following Lemma A.3. Perturb g to a β-regular, complete metric g', sufficiently C^2 close to g, such that the following conditions hold:

(1) $R^{\lambda_{g'}}$ is C^0 ϵ-close to R^{λ_g}.

(2) Let

$$\widetilde{U} = \{\gamma \in \mathcal{L}_{\widetilde{\beta}} S^*X \mid \ell g'_S(\gamma) < \sqrt{E'}\},$$

where

$$M = \max_{\gamma \in N_0} \ell_{g_S}(\gamma) < \sqrt{E'} < \sqrt{E}_0.$$

Then $\widetilde{U} \subset \widetilde{U}_{\sqrt{E_0}}$ and $\widetilde{U} \supset W$.

Thus $\widetilde{U}$ satisfies the conditions of the continuation Lemma A.4 and so

$$F(g, \beta) := i(N_0, \widetilde{\beta}) = i(N_1, \widetilde{\beta}).$$

where $N_1 = (\widetilde{U} \cap S(R^{\lambda_{g'}}, \widetilde{\beta}))/S^1$.

It thus remains to compute $i(N_1, \widetilde{\beta})$. Set

$$U^{g'}_{E'} = \{\gamma \in \mathcal{L}_\beta X \mid \mathcal{E}_{g'}(\gamma) < E'\}.$$

Since g' is β-regular, the critical set of $\mathcal{E}_{g'}$ in $U^{g'}_{E'}$ is a finite union of critical circles

$$C_o \simeq S^1,$$

one for each element of N_1.

The unstable manifolds of C_o for the g'-energy flow (negative gradient flow) on $\mathcal{L}_\beta X$ give a standard Morse-Bott handlebody $\mathcal{MB} \subset U^{g'}_{E'}$, that is invariant under the S^1 action. Standard Morse theory, (as we have already established Palais–Smale conditions) give that $\mathcal{MB} \to U^{g'}_{E'}$ is an S^1-equivariant homotopy equivalence.

After passing to the quotient by S^1, the unstable disk bundle over C_o becomes a cell of dimension morse(o). Therefore $\mathcal{MB}/S^1$ has a finite CW decomposition with one k-cell for each class-β g'-geodesic string $o \in U^{g'}_{E'}$ satisfying

$$\mathrm{morse}(o) = k.$$

For g' sufficiently C^2 nearby to g, $U^{g'}_{E'}$ is S^1-equivariantly homotopy equivalent to U_{E_0}. This is consequence of sandwich Lemma 7.2 just after the main proof, applied to $A = U_M$, $B = U_{E_0}$. Also U_{E_0} is S^1-equivariantly homotopic to $L_\beta X$ as was previously shown. Hence the finite CW decomposition above proves that $H^{S^1}_*(L_\beta X, \mathbb{Z})$ has finite total rank.

We then have:

$$\chi(\mathcal{MB}/S^1) = \chi(U^{g'}_{E'}/S^1) = \sum_{o \in N_1} (-1)^{\mathrm{morse}(o)}.$$

On the other hand, since g' is β-regular, the Fuller index formula gives

$$i(N_1, \widetilde{\beta}) = \sum_{o \in N_1} \frac{(-1)^{\mathrm{morse}(o)}}{\mathrm{mult}(o)}.$$

Since β is indivisible the denominators above are 1. And so

$$i(N_1, \widetilde{\beta}) = \chi(U^{g'}_{E'}/S^1) = \chi(U_{E_0}/S^1) = \chi(L_\beta X/S^1).$$

□

We state the following lemma separately for while it is very elementary it might be of independent interest.

Lemma 7.2. *For $a \in \mathbb{R}_{\geq 0}$ set*
$$S_a = \{\mathcal{E}_g \leq a\}.$$
Let $a < b$ and suppose that $\mathcal{E}_g$ satisfies the Palais–Smale condition on
$$\mathcal{B} = \{a \leq \mathcal{E}_g \leq b\}$$
and has no critical points on
$$\{a < \mathcal{E}_g \leq b\}.$$
Let g' be sufficiently C^2-close to g, so that $\nabla \mathcal{E}_{g'}$ is C^0-close to $\nabla \mathcal{E}_g$ on this band. Suppose
$$S_{a+\delta} \subset C \subset S_b$$
for some $\delta > 0$, where
$$C = \{\mathcal{E}_{g'} < c\}.$$
Then the inclusion
$$C \hookrightarrow B$$
is an S^1-equivariant homotopy equivalence.

Proof. By the Palais–Smale condition and the absence of critical points in $\mathcal{B} = \{a+\delta \leq \mathcal{E}_g \leq b\}$, there exists $\delta > 0$ such that on $\mathcal{B}$
$$\|\nabla \mathcal{E}_g\| \geq \delta. \tag{7.3}$$
For g' sufficiently close to g, we have
$$\|\nabla \mathcal{E}_{g'} - \nabla \mathcal{E}_g\| < \delta/2 \tag{7.4}$$
on $\mathcal{B}$. Therefore
$$\langle \nabla \mathcal{E}_g, \nabla \mathcal{E}_{g'} \rangle \geq \|\nabla \mathcal{E}_g\|^2 - \|\nabla \mathcal{E}_g\| \|\nabla \mathcal{E}_{g'} - \nabla \mathcal{E}_g\| > 0,$$
on $\mathcal{B}$. Thus along the negative $\mathcal{E}_{g'}$-gradient flow,
$$\frac{d}{dt}\mathcal{E}_g = -\langle \nabla \mathcal{E}_g, \nabla \mathcal{E}_{g'} \rangle < 0$$
on $\mathcal{B}$. Hence $\mathcal{B}$ is forward-invariant under this flow.

Moreover, $\mathcal{E}_{g'}$ is non-critical on $\mathcal{B}$ by (7.3), (7.4). Since $C \subset \mathcal{B}$, the standard deformation argument as in Palais [13, Section 10, Proposition (2)] therefore gives a deformation retraction of B onto C.

Finally, the energies and Hilbert metrics are S^1-invariant, so the gradient flow is S^1-equivariant. Hence the deformation retraction is S^1-equivariant. □

8. Proof of Theorem 1.31 and its corollaries

We first prove the following.

Theorem 8.1 (Product formula)**.** *Let*
$$Z \hookrightarrow (X, g) \xrightarrow{p} (Y, g_Y)$$
be a β-periodic fibration, with Y closed, connected and such that all smooth closed contractible g_Y-geodesics in Y are constant.

Fix $y_0 \in Y$, set $Z_0 = p^{-1}(y_0)$, and choose
$$\beta_Z \in \pi_1(Z_0)$$

such that

$$(i_{y_0})_*(\beta_Z) = \beta,$$

where $i_{y_0} : Z_0 \hookrightarrow X$ *is the inclusion. Denote by* g_{y_0} *the restriction of* g *to the fiber of* y_0*. Let* H_{y_0} *be the holonomy group of the fibration at* y_0*, acting on* $\pi_1^{\mathrm{inc}}(Z_{y_0})$*, Set*

$$S_{y_0} = H_{y_0} \cdot \beta_Z$$

be the holonomy orbit of β_Z*. Then*

$$F(g, \beta) = (-1)^{\dim Y} \cdot \# S_{y_0} \cdot \chi(Y) \cdot F(g_{y_0}, \beta_Z).$$

Proof. We first show that every class-β g-geodesic in X is vertical.

Let

$$o : S^1 \to X$$

be a closed g-geodesic representing the class β. Since β is a fiber class, the loop

$$p \circ o : S^1 \to Y$$

is contractible. By the definition of a β-periodic fibration, p is a Riemannian submersion and so projects g-geodesics to g_Y-geodesics. Hence $p \circ o$ is a smooth closed contractible g_Y-geodesic. By hypothesis, every such geodesic is constant. Therefore $p \circ o$ is constant, and so o is contained in a single fiber $Z_y = p^{-1}(y)$. Thus the class-β geodesic strings in X are precisely vertical geodesic strings.

Let

$$C = S^* X$$

be the unit cotangent bundle of (X, g), with projection

$$\mathrm{pr} : C \to X.$$

Let R^{λ_g} denote the Reeb vector field on C. Let $\widetilde{\beta} \in \pi_1(C)$ be the canonical lift of β. We want to compute:

$$F(g, \beta) = i\big(\mathcal{O}(R^{\lambda_g}, \widetilde{\beta}), R^{\lambda_g}, \widetilde{\beta}\big).$$

The idea is to localize contributions over critical points of a Morse function on Y.

Let

$$\pi^{\mathrm{vert}} : TX \to T^{\mathrm{vert}} X = \ker(dp)$$

be the g-orthogonal projection. By assumptions on $p : X \to Y$,

$$\nabla \pi^{\mathrm{vert}} = 0,$$

where π^{vert} is viewed as a $(1,1)$-tensor field.

For $\xi \in C$, let

$$v_\xi \in T_{\mathrm{pr}(\xi)} X$$

be the g-dual unit vector to ξ. Define

$$P : C \to \mathbb{R}$$

by

$$P(\xi) = |\pi^{\mathrm{vert}} v_\xi|_g^2.$$

Thus $P = 1$ on vertical unit covectors and $P = 0$ on horizontal unit covectors.

Let $f : Y \to \mathbb{R}$ be a Morse function. Define

$$\widetilde{f} : C \to \mathbb{R}$$

by

$$\widetilde{f} = f \circ p \circ \mathrm{pr}\,.$$

Equip $C = S^* X$ with the Sasaki metric g_S induced by g, as in the Proof of Theorem 1.24. We shall use the following two elementary identities.

First,

$$dP(R^{\lambda_g}) = 0. \tag{8.2}$$

Indeed, let $\xi(\tau)$ be a flow line of R^{λ_g}, and set

$$x(\tau) = \mathrm{pr}(\xi(\tau)).$$

Then $x(\tau)$ is a unit-speed g-geodesic. Let

$$v(\tau) = \dot{x}(\tau) = v_{\xi(\tau)}$$

be its velocity vector field. Since $x(\tau)$ is a geodesic,

$$\nabla_\tau v = 0.$$

Since $\nabla \pi^{\mathrm{vert}} = 0$, we have

$$\nabla_\tau(\pi^{\mathrm{vert}} v) = \pi^{\mathrm{vert}}(\nabla_\tau v) = 0.$$

Therefore

$$\begin{aligned} \frac{d}{d\tau} P(\xi(\tau)) &= \frac{d}{d\tau} |\pi^{\mathrm{vert}} v(\tau)|_g^2 \\ &= 2 \left\langle \nabla_\tau(\pi^{\mathrm{vert}} v), \pi^{\mathrm{vert}} v \right\rangle_g \\ &= 0. \end{aligned}$$

Hence $dP(R^{\lambda_g}) = 0$.

Second,

$$dP(\nabla_{g_S} \widetilde{f}) = 0. \tag{8.3}$$

Since $\widetilde{f} = f \circ p \circ \mathrm{pr}$ depends only on the base point $\mathrm{pr}(\xi)$:

$$\nabla_{g_S} \widetilde{f} \in T^{\mathrm{hor}} C.$$

More precisely, $\nabla_{g_S} \widetilde{f}$ is the horizontal lift of

$$\nabla_g (f \circ p)$$

on X. Let $\xi(s)$ be a horizontal curve in C, and set

$$x(s) = \mathrm{pr}(\xi(s)).$$

By definition of the Levi-Civita horizontal distribution on S^*X, the covectors $\xi(s)$ are parallel along $x(s)$. Equivalently, their g-dual vectors

$$v(s) = v_{\xi(s)}$$

are parallel along $x(s)$:

$$\nabla_s v(s) = 0.$$

Using again $\nabla \pi^{\mathrm{vert}} = 0$, we get

$$\nabla_s(\pi^{\mathrm{vert}} v(s)) = \pi^{\mathrm{vert}}(\nabla_s v(s)) = 0.$$

Hence

$$\begin{aligned} \frac{d}{ds} P(\xi(s)) &= \frac{d}{ds} |\pi^{\mathrm{vert}} v(s)|_g^2 \\ &= 2 \left\langle \nabla_s(\pi^{\mathrm{vert}} v(s)), \pi^{\mathrm{vert}} v(s) \right\rangle_g \\ &= 0. \end{aligned}$$

Taking $\dot{\xi}(0) = \nabla_{g_S} \widetilde{f}(\xi)$ gives

$$dP_\xi(\nabla_{g_S} \widetilde{f}) = 0.$$

Since ξ was arbitrary,

$$dP(\nabla_{g_S} \widetilde{f}) = 0.$$

Now set

$$V_\varepsilon = R^{\lambda_g} - \varepsilon \nabla_{g_S}(P + \widetilde{f}).$$

Recall first the discussion of Fuller continuation in the Appendix. Now

$$N_0 = \mathcal{O}(R^{\lambda_g}, \beta)$$

is compact by β-tautness, take $\varepsilon > 0$ to be small enough so that V_ϵ is C^0 ϵ'-close to R^{λ_g}, where ϵ' is the Fuller constant of N_0. The Fuller continuation of N_0 is a compact set $N_\varepsilon \subset S(V_\epsilon, \beta)$ and

$$i\big(N_0, R^{\lambda_g}, \widetilde{\beta}\big) = i(N_\varepsilon, V_\varepsilon, \widetilde{\beta}).$$

Note that the above does not require any further properties of V_ϵ, like completeness. We setup the Fuller index theory to sidesteps such issues.

We now identify the closed orbits in N_ε. We show that P is a Lyapunov function for V_ϵ. Mainly, along a trajectory $\xi(\tau)$ of V_ε, we have

$$\begin{aligned}\frac{d}{d\tau}P(\xi(\tau)) &= dP(R^{\lambda_g}) - \varepsilon dP(\nabla_{g_S}P) - \varepsilon dP(\nabla_{g_S}\widetilde{f})\\ &= -\varepsilon|\nabla_{g_S}P|^2_{g_S} \le 0.\end{aligned}$$

Thus P is nonincreasing along the V_ε-flow.

If $\xi(\tau)$ is a closed orbit of V_ε, then $P(\xi(\tau))$ returns to its initial value after one period. Since P is nonincreasing, it must be constant along the orbit. Hence

$$\nabla_{g_S}P = 0$$

along the orbit.

On each unit cotangent sphere, P is the squared norm of the vertical component of the corresponding unit vector. Therefore the critical locus of P consists of the two branches

$$P = 1 \qquad \text{and} \qquad P = 0,$$

corresponding respectively to vertical and horizontal covectors. At $\varepsilon = 0$, the class-$\widetilde{\beta}$ orbit set under consideration is vertical, hence lies in $P = 1$. For $\varepsilon > 0$ sufficiently small, the continued set N_ε remains in the continuation of this vertical component. Thus the horizontal branch $P = 0$ is excluded, and every closed orbit in N_ε lies in the vertical branch $P = 1$.

Consequently, every closed orbit in N_ε projects to a geodesic contained in a fiber Z_y.

It remains to see over which fibers these vertical orbits occur. On the vertical branch $P = 1$, the Reeb vector field restricts to the geodesic Reeb field of the fiber (Z_y, g_y), while $\widetilde{f}$ depends only on the base point y. The perturbation

$$-\varepsilon\nabla_{g_S}\widetilde{f}$$

moves the base point in the negative gradient direction of f. Therefore a vertical closed orbit of V_ε can occur only over a critical point of f. Conversely, if $y \in \mathrm{Crit}(f)$, then

$$\nabla_{g_S}\widetilde{f} = 0$$

over Z_y, and the vertical part of V_ε is just the fiber geodesic flow, up to the harmless P-term which vanishes on the vertical critical branch. Thus the closed orbits of V_ε in N_ε are precisely the vertical closed geodesic orbits lying in fibers over critical points of f.

We now compute the index. For $y \in Y$, let

$$H_y \subset \mathrm{Aut}\,\pi_1(Z_y)$$

be the holonomy group of the fibration at y. Choose

$$\beta_Z \in \pi_1(Z_y)$$

with

$$(i_y)_*(\beta_Z) = \beta.$$

Set

$$S_y = H_y \cdot \beta_Z.$$

Now the Fuller index is additive, that is N, N' are compact open and $N \cap N' = \emptyset$ then

$$i(N + N', V, \beta) = i(N, V, \beta) + i(N', V, \beta).$$

So it remain to compute the local index contribution $i(N_{\epsilon,y}, V_\epsilon, \widetilde{\beta})$ corresponding to the isolated subset of N_ϵ lying over a critical point y of f. We defer this computation to the Appendix as Lemma B.1; it yields that:

$$i(N_{\varepsilon,y}, V_\varepsilon, \widetilde{\beta}) = (-1)^{\dim Y + \mathrm{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha), \tag{8.4}$$

Next, we show that $F(g_y, \cdot)$ is constant on S_y. Indeed, let

$$\gamma : [0,1] \to Y$$

be a path based at y. Pulling back $p : X \to Y$ along γ gives a fibration over $[0,1]$. By the definition of a β-taut fibration, this pullback is a taut deformation of the fiber metric from (Z_y, g_y) back to itself, with final fiber identified with the initial one by the holonomy map $h_\gamma : Z_y \to Z_y$. Fuller index invariance under taut deformation gives

$$F(g_y, \alpha) = F(g_y, h_{\gamma,*}\alpha).$$

Thus $F(g_y, \alpha)$ is constant on the holonomy orbit S_y. Hence

$$\sum_{\alpha \in S_y} F(g_y, \alpha) = \# S_y \cdot F(g_y, \beta_Z).$$

If $y, y' \in Y$, holonomy along any path from y to y' identifies S_y with $S_{y'}$. The same path gives a taut deformation between the fiber metrics, so

$$\sum_{\alpha \in S_y} F(g_y, \alpha)$$

is independent of y. So we get:

$$\begin{aligned}
F(g, \beta) &= \sum_{y \in \mathrm{Crit}(f)} i(N_{\varepsilon,y}, V_\varepsilon, \widetilde{\beta}) \\
&= \sum_{y \in \mathrm{Crit}(f)} (-1)^{\dim Y + \mathrm{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha), \\
&= \sum_{y \in \mathrm{Crit}(f)} (-1)^{\dim Y + \mathrm{morse}(y)} \cdot \# S_{y_0} \cdot F(g_{y_0}, \beta_Z) \\
&= (-1)^{\dim Y} \cdot \chi(Y) \cdot \# S_{y_0} \cdot F(g_{y_0}, \beta_Z).
\end{aligned}$$

□

We return to the proof of Theorem 1.31. The first part immediately follows from the second, as any class $\beta \in \pi_1^{inc}(X)$ geodesic strings of a complete negatively curved Riemannian manifold X are unique. We prove second part. Suppose first that β is an n-power, so that β has a representative with multiplicity n, and we let α denote the corresponding indivisible class. By the assumption that all contractible g_Y geodesics are constant, classical Morse theory Milnor [12] tells us that Y has vanishing higher homotopy groups $\pi_k(Y, y_0)$, $k \geq 2$. And in particular $i_{Z,*} : \pi_1(Z, p_0) \to \pi_1(X, p_0)$ is a group injection, by the long exact sequence of a fibration. It follows that $\alpha \in \pi_1^{inc}(X)$ is also a fiber class.

Now, any α-class g-geodesic string must be contained in a fiber of p. For otherwise we may find a β class g-geodesic string, which is not contained in a fiber of p, which would contradict verticality in the first part of the proof of the product formula. It readily follows that $p: X \to Y$ is also α-taut.

Now, if $\chi(Y) \neq \pm 1$ then by the product formula $F(g,\alpha) \neq 1$, since $F(g_{y_0}, \alpha_Z)$ is an integer by Theorem 1.24. By Theorem 1.24

$$F(g,\alpha) = \chi^{S^1}(L_\alpha X).$$

Now, if X admits a complete metric g with a unique and non-degenerate class β g-geodesic string o_β, then it clearly also has a unique class α g-geodesic string o_α. Furthermore, o_α must be non-degenerate as otherwise o_β, which is a k-fold cover of o_α is degenerate. Then we have:

$$F(g,\alpha) = \chi^{S^1}(L_\alpha X) = 1$$

(see Proof of Theorem 1.24), which is impossible so we get a contradiction.

We now prove the general case. Suppose by contradiction that X admits a metric with a unique and non-degenerate class β g-geodesic string o_β. By the above, β is not atomic. Now o_β covers a multiplicity one geodesic string o_α in some class $\alpha \in \pi_1^{inc}(X)$. Moreover, o_α is the unique geodesic string in its class, otherwise o_β would not be unique in its class. We prove that α is indivisible, which will be a contradiction to β not being atomic and will complete the proof.

Suppose otherwise, so that $\alpha_{x_0} = \rho^k$ for $k > 1$ and $\rho \in \pi_1(X, x_0)$, (with the notation of Definition 1.22). Let o_ρ be a class ρ, closed g-geodesic string (where by slight abuse ρ also denotes the class in $\pi_1^{inc}(X)$ corresponding to the based class ρ.) It is immediate that the k cover of o_ρ, o_ρ^k represents α and is a g-geodesic string. By the uniqueness, $o_\alpha = o_\rho^k$. But this contradicts simplicity of o_α. So α is indivisible.

□

Proof of Theorem 1.6. Let g_Z, g_Y be complete Riemannian metrics on Z with g_Z having non-positive sectional curvature and g_Y not admitting contractible closed geodesics. Take the product metric $g = g_Z \times g_Y$ on $X = Z \times Y$. For a class $\beta \in \pi_1^{inc}(X)$ in the image of the inclusion $\pi_1^{inc}(Z) \to \pi_1^{inc}(X)$, the natural projection $X \to Y$ is automatically a β-taut fibration. Then the conclusion readily follows from Theorem 1.31.

□

Proof of Theorem 1.26. By Theorem 8.1 0 is certainly a value of the invariant F. We first prove that every negative rational number is the value of the invariant. Let p, q be positive integers. Let Y be a closed surface of genus $(p+1) > 1$ with a hyperbolic metric g_Y, let Z be the genus 2 closed surface with a hyperbolic metric g_Z and let $\beta_Z \in \pi_1^{inc}(Z)$ be the class represented by a $2 \cdot q$-fold covering of a simple closed loop representing a generator of the fundamental group of Z.

Let $X = Y \times Z$ with the product metric $g = g_Y \times g_Z$ and $p: X \to Y$ the canonical projection. By Theorem 8.1

$$F(g,\beta) = \chi(Y) \cdot F(g_Z, \beta_Z) = (-2p) \cdot \frac{1}{2q} = -\frac{p}{q},$$

where β is as in Lemma **??**. So we proved our first claim.

Let again p, q be positive integers. Let Y be closed surface of genus 2, with a hyperbolic metric g_Y. And let Z be a manifold satisfying $F(g_Z, \beta_Z) = -\frac{p}{2q}$ for some β_Z-taut metric g_Z on Z and for some class $\beta_Z \in \pi_1^{inc}(Z)$. This exist by the discussion above. Let $g = g_Y \times g_Z$ be the product metric on $Y \times Z$, and β as above. Analogously to the discussion above we get:

$$F(g,\beta) = \chi(Y) \cdot F(g_Z, \beta_Z) = (-2) \cdot \frac{-p}{2q} = \frac{p}{q}.$$

□

Appendix A. Fuller index and sky catastrophes

In this appendix we recall the part of Fuller's periodic orbit index used in the paper. We also fix the notation for orbit strings and for continuation under small perturbations of vector fields.

Let X be a smooth manifold, and fix a complete auxiliary Riemannian metric h on X. Let

$$\mathcal{V}(X)$$

denote the space of complete smooth nonsingular vector fields on X, with the C^0-topology on compact subsets.

For $V \in \mathcal{V}(X)$ and a free homotopy class $\beta \in \pi_1(X)$, define

$$S(V, \beta) = \{o \in L_\beta X \mid \dot{o}(t) = T\,V(o(t)) \text{ for some } T > 0\}\,.$$

The number T is uniquely determined by o, since V has no zeros. We call it the period of o, and write

$$T = T(o).$$

The circle $S^1 = \mathbb{R}/\mathbb{Z}$ acts on $S(V, \beta)$ by reparametrization:

$$(s \cdot o)(t) = o(t + s).$$

We define the space of orbit strings by

$$O(V, \beta) = S(V, \beta)/S^1.$$

When no confusion is possible, we use the same letter o for a parametrized orbit and for its associated orbit string.

Let

$$q : L_\beta X \to L_\beta X/S^1$$

be the quotient map. If

$$U \subset L_\beta X/S^1,$$

we write

$$\widehat{U} = q^{-1}(U) \subset L_\beta X.$$

We say that U is h-length bounded if the h-lengths of all loops in $\widehat{U}$ are uniformly bounded.

For an orbit string $o \in O(V, \beta)$, its multiplicity

$$m(o) \in \mathbb{N}$$

is the unique positive integer such that o is the $m(o)$-fold cover of a primitive orbit string. Equivalently, if $T(o)$ is the period of o and $T_{\mathrm{prim}}(o)$ is the period of the underlying primitive orbit string, then

$$m(o) = \frac{T(o)}{T_{\mathrm{prim}}(o)}.$$

A.1. **Fuller index.** We first recall the definition in the nondegenerate case. Suppose

$$N \subset O(V, \beta)$$

is finite and consists of nondegenerate orbit strings. Fuller associates to (N, V, β) the rational number

$$i(N, V, \beta) = \sum_{o \in N} \frac{i(o)}{m(o)}.$$

Here $i(o)$ is the fixed point index of the time-$T(o)$ return map of the flow of V, computed on a local hypersurface transverse to the image of o.

For a general compact open subset

$$N \subset O(V, \beta),$$

the Fuller index can be defined by first making a sufficiently small perturbation of V, supported in an isolating neighborhood of N, so that the corresponding orbit strings become nondegenerate. Fuller's construction shows that the resulting number is independent of the auxiliary perturbation.

In the special case of a nondegenerate closed geodesic orbit, the local fixed point index agrees with the usual Morse sign:

$$i(o) = (-1)^{\mathrm{morse}(o)}.$$

Consequently, for a regular geodesic flow, Fuller's index becomes

$$i(N, V, \beta) = \sum_{o \in N} \frac{(-1)^{\mathrm{morse}(o)}}{m(o)}.$$

A.2. **Continuation.** Let

$$\{V_t\}_{t \in [0,1]} \subset \mathcal{V}(X)$$

be a continuous family. Define

$$S(\{V_t\}, \beta) = \{(o, t) \in L_\beta X \times [0,1] \mid o \in S(V_t, \beta)\},$$

and

$$O(\{V_t\}, \beta) = S(\{V_t\}, \beta)/S^1, \tag{A.1}$$

whereS^1 acts on the loop coordinate.

Fuller's basic invariance says the following. Let

$$\mathcal{N} \subset O(\{V_t\}, \beta)$$

be compact and open. Suppose that its endpoint slices are

$$N_0 = \mathcal{N} \cap O(V_0, \beta), \qquad N_1 = \mathcal{N} \cap O(V_1, \beta).$$

Then

$$i(N_0, V_0, \beta) = i(N_1, V_1, \beta). \tag{A.2}$$

We refer to this as the basic invariance of the Fuller index.

For applications, we need a local version of this statement under small perturbations. This is a variation of [8, Lemma 4.1], with the added complexity needed for dealing with noncompact X.

Lemma A.3 (Isolation lemma)**.** *Let h be a complete auxiliary Riemannian metric on X. Let V_0 be a smooth nonsingular vector field on X, and let*

$$N_0 \subset O(V_0, \beta)$$

be compact and open. Let

$$U \subset L_\beta X/S^1$$

be an open neighborhood of N_0 such that

$$\overline{U} \cap O(V_0, \beta) = N_0.$$

Assume that U is controlled: there are a compact set $K \subset X$ and a constant $L > 0$ such that every loop $o \in \widehat{U}$ satisfies

$$\mathrm{image}(o) \subset K, \qquad \ell_h(o) \leq L.$$

Then for every open neighborhood W of N_0 satisfying

$$\overline{W} \subset U,$$

there exists $\epsilon > 0$ with the following property. If V is a smooth vector field such that

$$\sup_{x \in K} |V(x) - V_0(x)|_h < \epsilon,$$

then

$$O(V, \beta) \cap U \subset W.$$

In particular,

$$O(V, \beta) \cap U$$

is compact open.

Proof. Since V_0 is nonsingular and K is compact, there are constants

$$0 < B < A < \infty$$

such that

$$B \leq |V_0(x)|_h \leq A$$

for all $x \in K$. We shall choose $\epsilon < B/2$. Then every vector field V satisfying

$$\sup_K |V - V_0|_h < \epsilon$$

obeys

$$|V(x)|_h \geq B/2, \qquad |V(x)|_h \leq A + \epsilon$$

for all $x \in K$.

Suppose the conclusion failed for the chosen W. Then there would be a sequence of smooth vector fields

$$V_j \to V_0 \qquad \text{uniformly on } K,$$

and orbit strings

$$o_j \in \big(O(V_j, \beta) \cap U\big) - W.$$

Choose parametrized representatives, still denoted o_j, satisfying

$$\dot{o}_j(s) = T_j V_j(o_j(s)).$$

Since $o_j \in \widehat{U}$, controlledness gives

$$\mathrm{image}(o_j) \subset K, \qquad \ell_h(o_j) \leq L.$$

For j sufficiently large, $|V_j|_h \geq B/2$ on K. Hence

$$\ell_h(o_j) = T_j \int_0^1 |V_j(o_j(s))|_h \, ds \geq T_j B/2.$$

Thus

$$T_j \leq \frac{2L}{B}.$$

Similarly, for j sufficiently large,

$$|V_j|_h \leq A + \epsilon$$

on K, and therefore

$$|\dot{o}_j(s)|_h = T_j |V_j(o_j(s))|_h \leq \frac{2L(A+\epsilon)}{B}.$$

Thus the o_j are equicontinuous and have images in the fixed compact set K. By Arzelà–Ascoli, after passing to a subsequence,

$$o_j \to o_\infty$$

uniformly. Also, after passing to a further subsequence, the periods converge:

$$T_j \to T_\infty.$$

Since $V_j \to V_0$ uniformly on K, passing to the limit in

$$\dot{o}_j = T_j V_j(o_j)$$

shows that

$$\dot{o}_\infty = T_\infty V_0(o_\infty).$$

Moreover $T_\infty > 0$, since

$$\ell_h(o_j) \geq T_j B/2$$

and the orbits are nonconstant in the class β. Hence o_∞ is a V_0-orbit in class β.

Now $o_\infty \in \overline{U} - W$. This contradicts

$$\overline{U} \cap O(V_0, \beta) = N_0 \subset W.$$

Therefore

$$O(V, \beta) \cap U \subset W$$

for all V sufficiently C^0-close to V_0 on K.

Note that $O(V,\beta) \cap U = O(V,\beta) \cap \overline{W}$, hence $O(V,\beta) \cap U$ is closed. The compactness of $O(V,\beta) \cap U$ then follows from the same argument. Every sequence of orbit strings in this set has representatives with image in K, length bounded by L, period bounded by $2L/B$, and uniformly bounded velocity. Thus Arzelà–Ascoli gives a convergent subsequence, and the limit is again an orbit string lying in $O(V,\beta) \cap U$. □

We will call ϵ above the ***Fuller constant*** of V_0, N_0, U, W (a priori it also depends on K, L). If we just say Fuller constant this data is implicitly chosen.

Lemma A.4 (Fuller continuation)**.** *Let V_0, N_0, U, W, K be as in Lemma A.3, and let ϵ be the associated Fuller constant. Let V_1 be ϵ C^0-close to V_0 on K, and set*

$$V_t = (1-t)V_0 + tV_1.$$

Then

$$\mathcal{N} = (U \times [0,1]) \cap O(\{V_t\}, \beta)$$

is compact and open in $O(\{V_t\}, \beta)$. Its endpoint slices are

$$N_0 = U \cap O(V_0, \beta), \qquad N_1 = U \cap O(V_1, \beta).$$

Consequently,

$$i(N_0, V_0, \beta) = i(N_1, V_1, \beta).$$

We call N_1 the **Fuller continuation** *of N_0.*

Proof. $(U \times [0,1]) \cap O(\{V_t\}, \beta)$ is clearly open, it is also closed since by Lemma A.3

$$(U \times [0,1]) \cap O(\{V_t\}, \beta) = (\overline{W} \times [0,1]) \cap O(\{V_t\}, \beta).$$

The compactness follows by the same argument as in Lemma A.3. The endpoint slices are plainly

$$\mathcal{N} \cap O(V_0, \beta) = U \cap O(V_0, \beta) = N_0$$

and

$$\mathcal{N} \cap O(V_1, \beta) = U \cap O(V_1, \beta) = N_1.$$

Fuller's basic invariance applied to the compact open cobordism $\mathcal{N}$ gives

$$i(N_0, V_0, \beta) = i(N_1, V_1, \beta).$$

□

Example 4 (Geodesic flows). Let (Y, g_Y) be a complete Riemannian manifold and let

$$C = S^*Y$$

be its unit cotangent bundle with the Liouville contact form λ_{g_Y}. The Reeb vector field $R^{\lambda_{g_Y}}$ is complete and nonsingular on C.

If $\beta \in \pi_1^{\mathrm{inc}}(Y)$, let

$$\widetilde{\beta} \in \pi_1(C)$$

be the canonical lift of β, obtained by lifting a nonzero constant-speed representative to the unit cotangent bundle.

Let g_S be the Sasaki metric on S^*Y. Thus

$$TC = T^{\mathrm{vert}}C \oplus T^{\mathrm{hor}}C$$

is an orthogonal splitting, where

$$T^{\mathrm{vert}}C = \ker(\mathrm{pr}_*)$$

and $T^{\mathrm{hor}}C$ is the horizontal subbundle induced by the Levi-Civita connection of g. And on $T^{\mathrm{hor}}C$, $g_S(v, w) = g(\pi_* v, \pi_* w)$ for $\pi_* : TC \to TX$ induced by the natural projection.

The basic property of this is that if o is a non-zero constant speed g-geodesic o, o is a constant-speed g_Y-geodesic and $\widetilde{o}$ is its unit cotangent lift, then

$$\ell_{g_S}(\widetilde{o}) = \ell_{g_Y}(o). \tag{A.5}$$

If $U \subset S(g_Y, \beta)$ is g_Y-length bounded, then the set of canonical lifts

$$\widetilde{U} \subset S(R^{\lambda_{g_Y}}, \widetilde{\beta})$$

is g_S-length bounded, by (A.5).

Appendix B. A local fibered calculation of F

We prove in this section the technical lemma computing the local index, used in Theorem 8.1.

Let

$$P : S^*X \to \mathbb{R}$$

be the verticality function as in the proof of Theorem 8.1,

$$P(\xi) = |\pi^{\mathrm{vert}} v_\xi|_g^2.$$

Here v_ξ is the g-dual unit vector to ξ, and

$$\pi^{\mathrm{vert}} : TX \to T^{\mathrm{vert}}X$$

is the g-orthogonal projection. The notation β, $\widetilde{\beta}$, and β_Z is likewise as in Theorem 8.1.

Lemma B.1 (Local product calculation)**.** *Let*

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a parallel Riemannian submersion. Let

$$\widetilde{f} = f \circ p \circ \mathrm{pr}$$

for a Morse function $f : Y \to \mathbb{R}$, and define the vector field on S^*X

$$V_\varepsilon = R^{\lambda^g} - \varepsilon \nabla_{g_S}(P + \widetilde{f}).$$

Fix

$$y \in \mathrm{Crit}(f),$$

and let

$$N_{\varepsilon,y}$$

be the isolated set of closed V_ε-orbit strings lying over the fiber

$$Z_y = p^{-1}(y),$$

as in the proof of Theorem 8.1. Then

$$i(N_{\varepsilon,y}, V_\varepsilon, \widetilde{\beta}) = (-1)^{\dim Y + \mathrm{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha),$$

where S_y is the holonomy orbit of β_Z. In particular, if the fibration is holonomy-finite in the relevant class, the sum is finite.

Proof. Let

$$T^{\mathrm{vert}}X = \ker(dp), \qquad H = (T^{\mathrm{vert}}X)^{\perp_g}.$$

Since the submersion is parallel, the two distributions

$$T^{\mathrm{vert}}X \quad \text{and} \quad H$$

are preserved by the Levi-Civita connection. Since $T^{\mathrm{vert}}X$ and $H = (T^{\mathrm{vert}}X)^{\perp_g}$ are parallel complementary distributions, the local de Rham decomposition theorem gives, gives a product decomposition

$$p^{-1}(D_y) \cong Z_y \times D_y, \qquad g = g_y \oplus g_Y,$$

for a normal ball neighborhood $D_y \subset Y$ centered at y. See Kobayashi–Nomizu [9, Chapter IV].

Set

$$C = S^*X,$$

and let

$$C^{\mathrm{vert}} = \{P = 1\} \subset C$$

be the vertical locus. Over Z_y, this locus is naturally identified with

$$S^*Z_y \subset S^*X.$$

Near

$$S^*Z_y \subset C^{\mathrm{vert}},$$

the product decomposition gives the normal-bundle model

$$S^*Z_y \times D_y \times B^*_y, \tag{B.2}$$

where

$$B^*_y \subset T^*_y Y$$

is a small ball in the horizontal cotangent direction. In this model, the vertical locus is

$$S^*Z_y \times D_y \times \{0\}.$$

A covector near the vertical locus decomposes as

$$\xi = \xi_{\mathrm{vert}} + \eta, \qquad \eta \in B^*_y.$$

The unit condition is

$$|\xi_{\mathrm{vert}}|^2_{g_y^*} + |\eta|^2_{g_Y^*} = 1.$$

Therefore

$$P(\xi) = |\xi_{\mathrm{vert}}|^2_{g_y^*} = 1 - |\eta|^2_{g_Y^*}.$$

Thus P has a Morse–Bott maximum along C^{vert}.

In the local model (B.4),

$$N_{\varepsilon,y} = N_y^{\mathrm{fib}} \times \{y\} \times \{0\},$$

where N_y^{fib} is the relevant closed orbit set of the fiber geodesic Reeb flow on S^*Z_y.

We now describe the local return map at an element of $N_{\varepsilon,y}$. Along the vertical locus over Z_y, the fiber component of V_ε is the geodesic Reeb field $R^{\lambda_{g_y}}$ on S^*Z_y. The base component is

$$-\varepsilon \nabla_{g_Y} f$$

on D_y. The normal covector component is the outward radial component coming from

$$-\varepsilon \nabla_{g_S} P$$

on B_y^*.

Let

$$o \in N_y^{\mathrm{fib}},$$

and choose a product transverse section

$$\Sigma = \Sigma^{\mathrm{fib}} \times D_y \times B_y^*,$$

where

$$\Sigma^{\mathrm{fib}} \subset S^*Z_y$$

is transverse to the fiber Reeb flow near o. With respect to this product section, the local return map of V_ε is the product

$$\Phi_{\mathrm{fib}} \times \psi_y \times \kappa.$$

Here:

- Φ_{fib} is the local return map of the fiber geodesic flow generated by $R^{\lambda_{g_y}}$;
- ψ_y is the small-time map of
$$-\varepsilon \nabla_{g_Y} f$$
near y;
- κ is the normal return map induced by
$$-\varepsilon \nabla_{g_S} P$$
near $0 \in B_y^*$.

The fixed point index is multiplicative under products, Dold [5, Chapter VII, Section 5]. Hence

$$i_{\mathrm{fp}}(o; V_\varepsilon) = i_{\mathrm{fp}}(o; R^{\lambda_{g_y}}) \cdot \mathrm{ind}_{PH}(-\nabla_{g_Y} f, y) \cdot \mathrm{ind}_{PH}(\kappa, 0).$$

Since y is a Morse critical point,

$$\mathrm{ind}_{PH}(-\nabla_{g_Y} f, y) = (-1)^{\mathrm{morse}(y)}.$$

Since the normal component of

$$-\varepsilon \nabla_{g_S} P$$

has outward radial linearization on a vector space of dimension $\dim Y$,

$$\mathrm{ind}_{PH}(\kappa, 0) = (-1)^{\dim Y}.$$

Therefore

$$i_{\mathrm{fp}}(o; V_\varepsilon) = (-1)^{\dim Y + \mathrm{morse}(y)} i_{\mathrm{fp}}(o; R^{\lambda_{g_y}}).$$

The multiplicity of o is unchanged by taking the product with the isolated base and normal fixed points. Thus

$$\frac{i_{\mathrm{fp}}(o;V_\varepsilon)}{m(o)} = (-1)^{\dim Y + \mathrm{morse}(y)} \frac{i_{\mathrm{fp}}(o;R^{\lambda_{g_y}})}{m(o)}.$$

Summing over all fiber orbit strings in the local block gives

$$i(N_{\varepsilon,y}, V_\varepsilon, \widetilde{\beta}) = (-1)^{\dim Y + \mathrm{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha).$$

This proves the claim. □

Let

$$P : S^* X \to \mathbb{R}$$

be the verticality function as in the proof of Theorem 8.1,

$$P(\xi) = |\pi^{\mathrm{vert}} v_\xi|_g^2.$$

Here v_ξ is the g-dual unit vector to ξ, and

$$\pi^{\mathrm{vert}} : TX \to T^{\mathrm{vert}} X$$

is the g-orthogonal projection. The notation β, $\widetilde{\beta}$, and β_Z is likewise as in Theorem 8.1.

Lemma B.3 (Local product calculation)**.** *Let*

$$Z \hookrightarrow X \xrightarrow{p} Y$$

be a parallel Riemannian submersion. Let

$$\widetilde{f} = f \circ p \circ \mathrm{pr}$$

for a Morse function $f : Y \to \mathbb{R}$, and define the vector field on S^*X

$$V_\varepsilon = R^{\lambda^g} - \varepsilon \nabla_{g_S}(P + \widetilde{f}).$$

Fix

$$y \in \mathrm{Crit}(f),$$

and let

$$N_{\varepsilon,y}$$

be the isolated set of closed V_ε-orbit strings lying over the fiber

$$Z_y = p^{-1}(y),$$

as in the proof of Theorem 8.1. Then

$$i(N_{\varepsilon,y}, V_\varepsilon, \widetilde{\beta}) = (-1)^{\dim Y + \mathrm{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha),$$

where S_y is the holonomy orbit of β_Z. In particular, if the fibration is holonomy-finite in the relevant class, the sum is finite.

Proof. Let

$$T^{\mathrm{vert}} X = \ker(dp), \qquad H = (T^{\mathrm{vert}} X)^{\perp_g}.$$

Since the submersion is parallel, the two distributions

$$T^{\mathrm{vert}} X \quad \text{and} \quad H$$

are preserved by the Levi-Civita connection. By the local de Rham product theorem, after shrinking a normal ball

$$D_y \subset Y$$

around y, there is a local Riemannian product decomposition

$$p^{-1}(D_y) \cong Z_y \times D_y$$

with

$$g = g_y \oplus g_Y.$$

In this decomposition, the fibers are the Z_y-factors and the horizontal leaves are the D_y-factors.

Let

$$C = S^*X.$$

Let

$$C^{\mathrm{vert}} = \{P = 1\} \subset C$$

be the vertical locus. Over Z_y, this locus is naturally identified with

$$S^*Z_y \subset S^*X.$$

Near

$$S^*Z_y \subset C^{\mathrm{vert}},$$

the product decomposition gives the normal-bundle model

$$S^*Z_y \times D_y \times B_y^*, \tag{B.4}$$

where

$$B_y^* \subset T_y^*Y$$

is a small ball in the horizontal cotangent direction. In this model, the vertical locus is

$$S^*Z_y \times D_y \times \{0\}.$$

A covector near the vertical locus decomposes as

$$\xi = \xi_{\mathrm{vert}} + \eta, \qquad \eta \in B_y^*.$$

The unit condition is

$$|\xi_{\mathrm{vert}}|^2_{g_y^*} + |\eta|^2_{g_Y^*} = 1.$$

Therefore

$$P(\xi) = |\xi_{\mathrm{vert}}|^2_{g_y^*} = 1 - |\eta|^2_{g_Y^*}.$$

Thus P has a Morse–Bott maximum along

$$C^{\mathrm{vert}},$$

and the normal component of

$$-\varepsilon \nabla_{g_S} P$$

has outward radial linearization along $\eta = 0$. Consequently the normal fixed point index contribution is

$$(-1)^{\dim Y}.$$

By the verticality identities for a parallel submersion,

$$dP(R^{\lambda^g}) = 0, \qquad dP(\nabla_{g_S}\widetilde{f}) = 0.$$

Hence, along a trajectory of V_ε,

$$\frac{d}{d\tau}P = -\varepsilon |\nabla_{g_S} P|^2_{g_S} \le 0.$$

On a closed orbit, P must return to its initial value, so

$$\nabla_{g_S} P = 0$$

along the orbit. Since $N_{\varepsilon,y}$ is the local continuation of the vertical orbit set, the horizontal critical locus of P is excluded. Therefore every closed orbit in $N_{\varepsilon,y}$ lies in the vertical locus.

In the local model (B.4), this gives

$$N_{\varepsilon,y} = N_y^{\mathrm{fib}} \times \{y\} \times \{0\},$$

where N_y^{fib} is the relevant closed orbit set of the fiber geodesic Reeb flow on S^*Z_y.

We now describe the local return map at an element of $N_{\varepsilon,y}$. Along the vertical locus over Z_y, the fiber component of V_ε is the geodesic Reeb field $R^{\lambda_{g_y}}$ on S^*Z_y. The base component is

$$-\varepsilon\nabla_{g_Y} f$$

on D_y. The normal covector component is the outward radial component coming from

$$-\varepsilon\nabla_{g_S} P$$

on B_y^*.

Let

$$o \in N_y^{\mathrm{fib}},$$

and choose a product transverse section

$$\Sigma = \Sigma^{\mathrm{fib}} \times D_y \times B_y^*,$$

where

$$\Sigma^{\mathrm{fib}} \subset S^* Z_y$$

is transverse to the fiber Reeb flow near o. With respect to this product section, the local return map of V_ε is the product

$$\Phi_{\mathrm{fib}} \times \psi_y \times \kappa.$$

Here:

- Φ_{fib} is the local return map of the fiber geodesic flow generated by $R^{\lambda_{g_y}}$;
- ψ_y is the small-time map of
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near y;
- κ is the normal return map induced by
$$-\varepsilon\nabla_{g_S} P$$
near $0 \in B_y^*$.

The fixed point index is multiplicative under products, Dold [5, Chapter VII, Section 5]. Hence

$$i_{\mathrm{fp}}(o, V_\varepsilon) = i_{\mathrm{fp}}(o, R^{\lambda_{g_y}}) \cdot \mathrm{ind}_{PH}(-\nabla_{g_Y} f, y) \cdot \mathrm{ind}_{PH}(\kappa, 0).$$

Since y is a Morse critical point,

$$\mathrm{ind}_{PH}(-\nabla_{g_Y} f, y) = (-1)^{\mathrm{morse}(y)}.$$

Since the normal component of

$$-\varepsilon\nabla_{g_S} P$$

has outward radial linearization on a vector space of dimension $\dim Y$,

$$\mathrm{ind}_{PH}(\kappa, 0) = (-1)^{\dim Y}.$$

Therefore

$$i_{\mathrm{fp}}(o, V_\varepsilon) = (-1)^{\dim Y + \mathrm{morse}(y)} i_{\mathrm{fp}}(o, R^{\lambda_{g_y}}).$$

The multiplicity of o is unchanged by taking the product with the isolated base and normal fixed points. Thus

$$\frac{i_{\mathrm{fp}}(o, V_\varepsilon)}{m(o)} = (-1)^{\dim Y + \mathrm{morse}(y)} \frac{i_{\mathrm{fp}}(o; R^{\lambda_{g_y}})}{m(o)}.$$

Summing over all fiber orbit strings in the local block gives

$$i(N_{\varepsilon,y}, V_\varepsilon, \widetilde{\beta}) = (-1)^{\dim Y + \mathrm{morse}(y)} \sum_{\alpha \in S_y} F(g_y, \alpha).$$

This proves the claim. $\square$

Appendix C. Sky catastrophes

We now recall the topological formulation of sky catastrophes. This is the version used in Fuller's theory and in the applications above.

Definition C.1. *Let*

$$\{V_t\}_{t\in[0,1]} \subset \mathcal{V}(X)$$

be a continuous family. We say that the family has a sky catastrophe in class β if there is an endpoint orbit string

$$y \in O(V_0,\beta) \cup O(V_1,\beta) \subset O(\{V_t\},\beta)$$

such that no compact open subset of $O(\{V_t\},\beta)$ contains y.

Equivalently, the endpoint orbit string y cannot be included in any compact continuation block through the family.

For suitably regular families, this agrees with the preliminary Definition 1.18.

Appendix D. Acknowledgements

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